

Zachary Gameiro

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TECHNICAL SKILLS

Programming: C/C++, Python, JavaScript/TypeScript, SQL, TI-BASIC

Testing & Environments: Jest, Mocha, Postman, endpoint testing, **Firebase Emulator Suite**, local/test/dev/staging/prod environment management

Systems & Validation: Git, Linux, CLI workflows, debugging, validation, file I/O, repeatable automation scripts

Embedded & Communications: Arduino, embedded systems, RF modules, wireless communication, serial/TRS interfaces, packet framing/retries

Automation & Tooling: Firebase, Supabase Edge Functions, Python pipelines, Docker, Google Cloud Run, structured extraction, schema validation

Languages: English, French, Portuguese

EDUCATION

Simon Fraser University

Bachelor of Science, Joint Major in Computing Science and Microbiology & Biochemistry

Burnaby, BC

Expected May 2027

SYSTEMS & TEST-ORIENTED PROJECTS

TI-84 Wireless Messaging — Personal Embedded Project

2020

Arduino C/C++, TI-BASIC, nRF24L01, embedded systems | Demo

- Reverse-engineered TI-84 link behavior and built a bridge between the calculator UI and Arduino C/C++ firmware for wireless messaging.
- Integrated RF hardware and implemented **packet framing**, retries, and parsing to improve communication reliability across devices.
- Debugged power stability, timing, and enclosure constraints during bring-up and validation of the system.

Haulr — Personal Project (Independent Build)

2026

React, TypeScript, Firebase, Cloud Functions, Stripe, Mapbox | Live Demo

- Built and maintained a multi-app operations platform with Firebase-backed workflows for quoting, scheduling, billing, and job execution.
- Used **Jest/Mocha, Postman**, and the **Firebase Emulator Suite** to validate function behavior, endpoint flows, and edge cases before deployment.
- Kept **local/test/dev/staging/prod** environments separated during backend rebuild work so changes could be verified safely before rollout.

Receipt OCR — Personal Project

2025

Python, PaddleOCR, Docker, Google Cloud Run | GitHub

- Built a **Python** pipeline that converts noisy receipt images into structured transaction JSON through OCR, layout reconstruction, and parsing.
- Reworked the extraction path to improve throughput by roughly **200%** versus the earlier prototype.
- Added normalization, validation, and edge-case handling so outputs remained reliable across varied receipt formats.

ACADEMIC PROJECTS

Ferry Terminal Booker — School Team Project

2025

C, File I/O, Disk Storage, CLI | GitHub

- Co-built a C-based reservation system with flat-file persistence, search flows, payment handling, and occupancy checks.
- Implemented safe read/modify/write flows and validation logic to keep stored records consistent without a database.

SMLM Clustering with Graph Embeddings — School Team Project

2025

Python, PyTorch, MIRO, DBSCAN, HDBSCAN | GitHub

- Co-built a Python/PyTorch data pipeline that used graph embeddings and density-based clustering on noisy microscopy data.
- Worked in a team setting to evaluate outputs, compare runs, and improve the overall analysis workflow.

EXPERIENCE

Sherwood Park Toyota — Online Sales & Marketing

May 2024 – Aug 2024

- Managed customer communication and documentation workflows in a high-volume environment.
- Identified a repetitive process problem and built internal automation to improve consistency and reduce manual work.

BCLDB — Consultant

Aug 2025 – Present

- Worked in a process-heavy retail environment requiring accuracy, verification, and dependable execution.
- Coordinated with teammates while following policy constraints and maintaining consistent service under volume.